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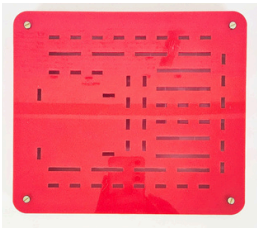
SESSION NAME

Session

6

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1 

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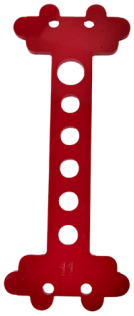
11 

12

straw

1

Spacers

2

L Clamp

3

Spacer Screws

4

Nuts

5

Screw

6

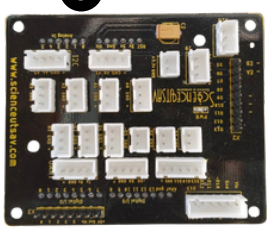
Washer

7

zinc wire 1M

8

screw hook

9 

Arduino With Shield

10

Arduino cable

ELECTRONIC COMPONENTS

Touch sensor

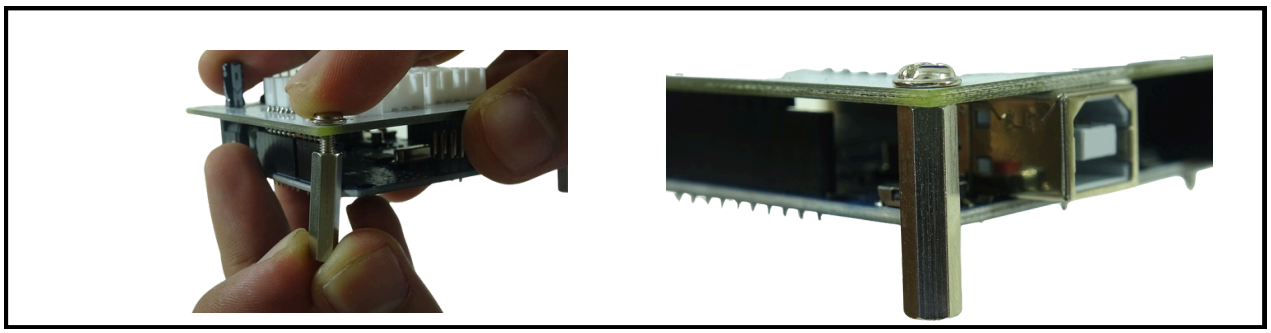
A touch sensor is an electronic device that detects human touch and works like a switch. It senses the change in capacitance when a finger touches it and sends a signal to the microcontroller. This signal can be used to perform actions like turning on an LED or activating a device. Touch sensors are commonly used in mobile screens, touch lamps, and smart systems.

2 PIN RMC

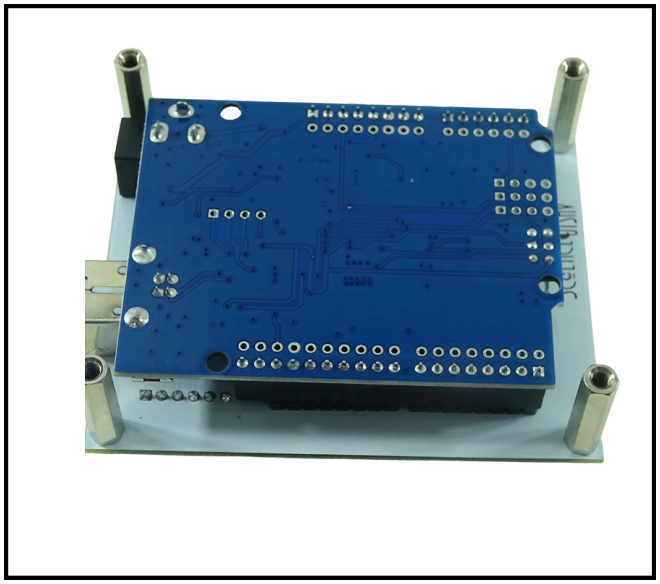
Used to connect the 1-watt LEDs and the buzzer.

Procedure :

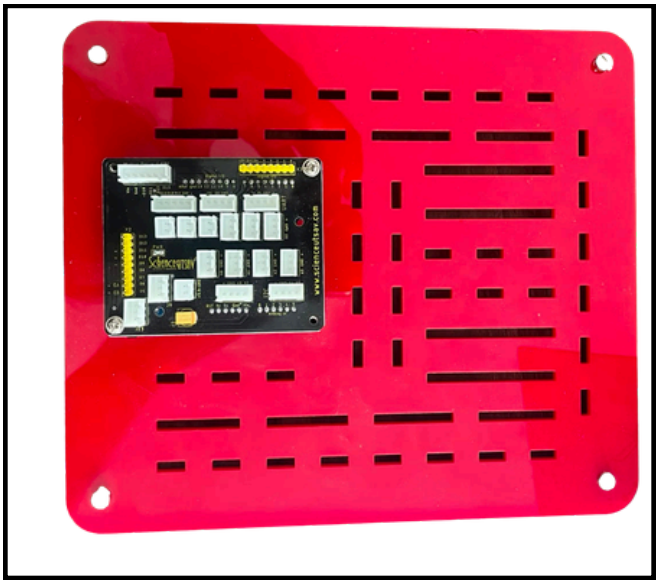
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



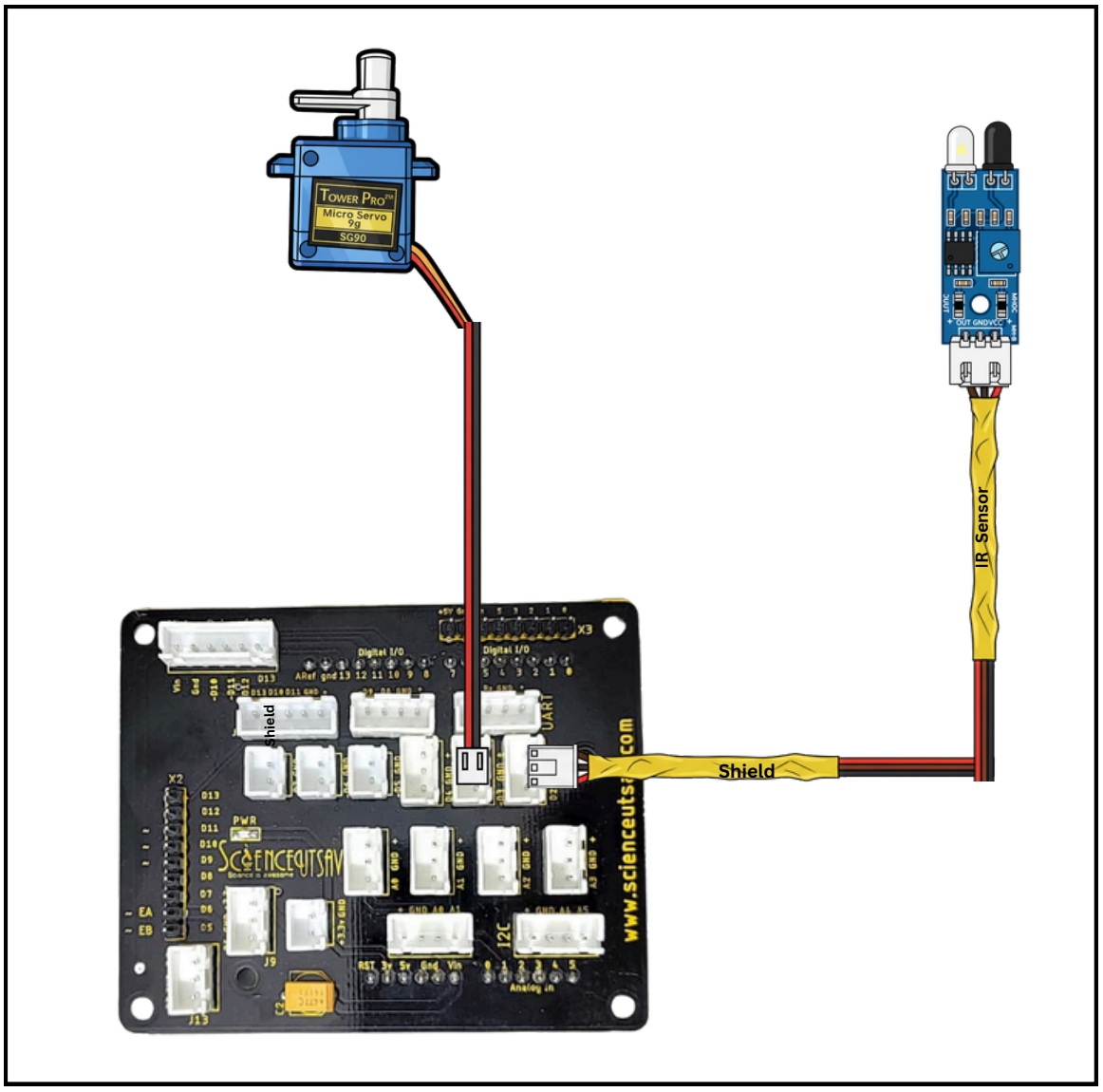
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

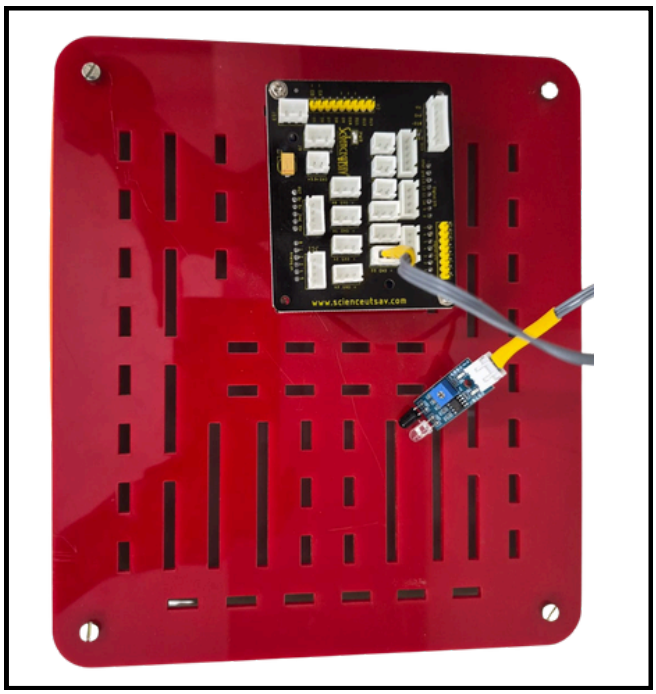
The circuit diagram of the project is given below



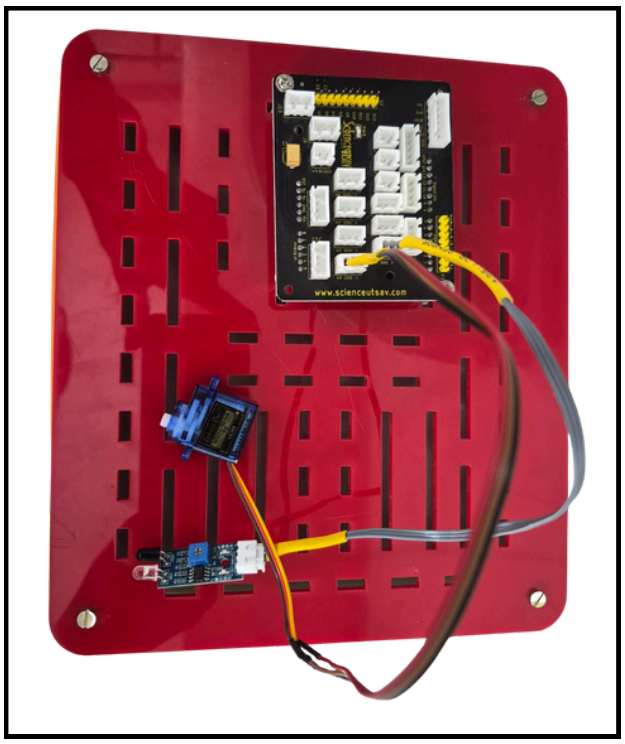
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

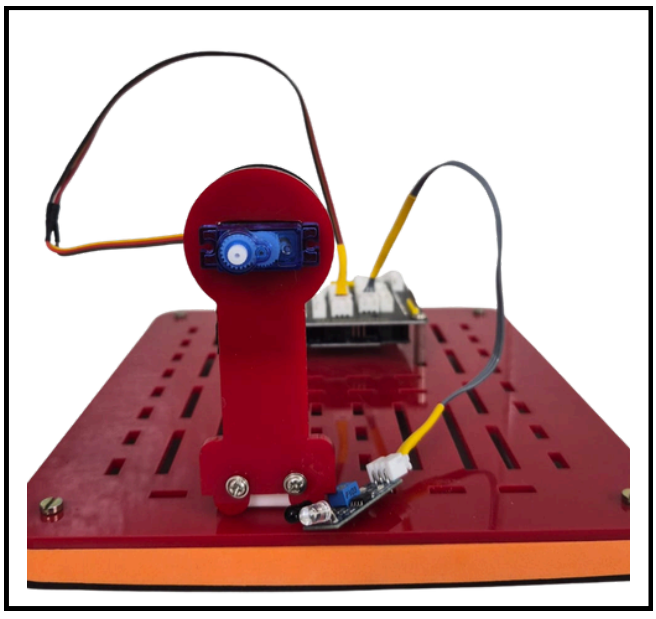
Step 1- Connect the IR sensor to the board



Step 2- Attach the servo motor to the board

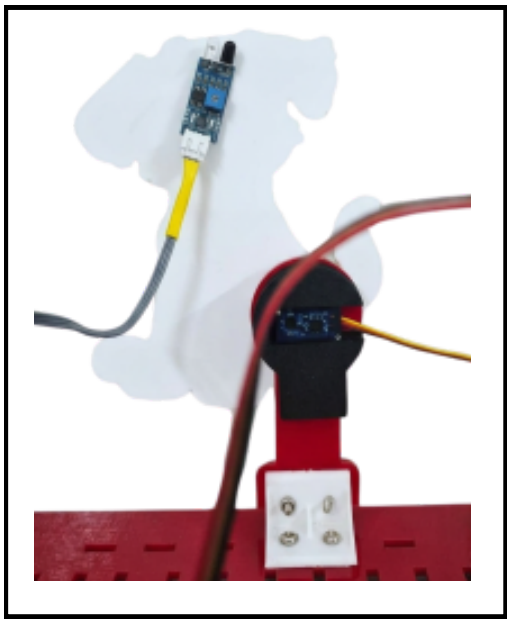


Step 3- Attach a stand to the base and insert the servo motor in it



Step 4- Screw in the template and connect it's tail to the servo motor

Step 5- insert the 3mm bolt to template

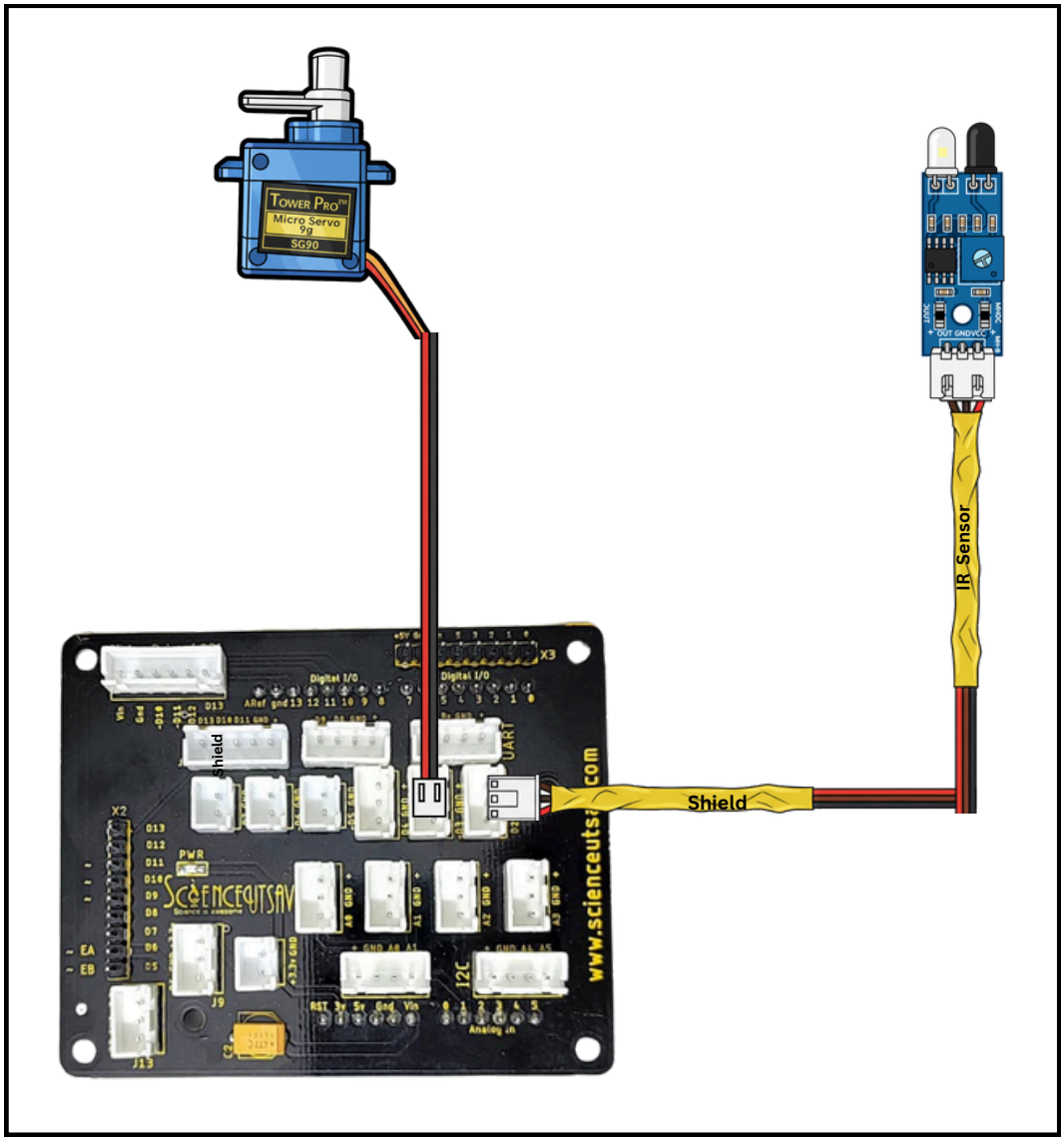


Code

```
// Session 6 Grade 6 Project 1
 // Dog Tail Movement
 // IR detected = dog tail moves

 #include <Servo.h>

 const int irSensor = 2;
 const int servoPin = 3;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 Servo tailServo;

 void setup() {
 pinMode(irSensor, INPUT);
 tailServo.attach(servoPin);
 tailServo.write(90);
 }

 void loop() {
 if (digitalRead(irSensor) == IR_DETECTED) {
   for (int i = 0; i < 3; i++) {
     tailServo.write(40);
     delay(300);
     tailServo.write(140);
     delay(300);
   }
   tailServo.write(90);
 }
 }
```

BEGINNER PROJECT

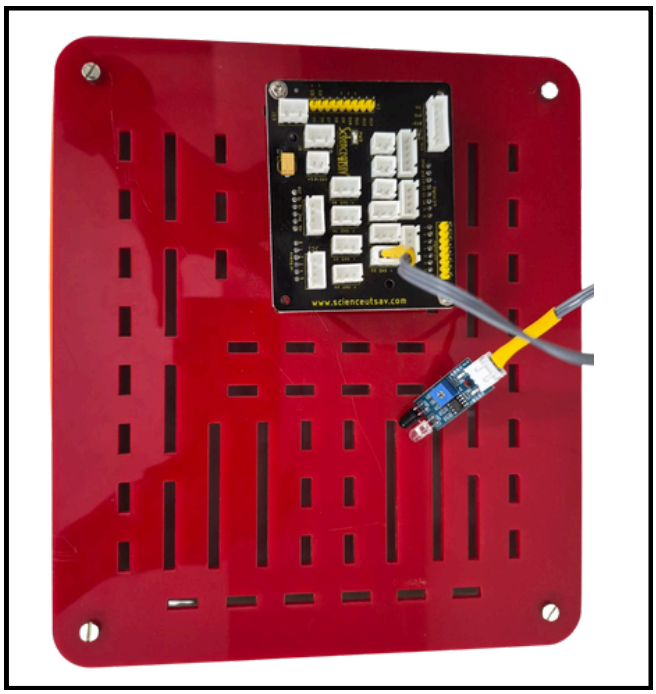
The circuit diagram of the project is given below



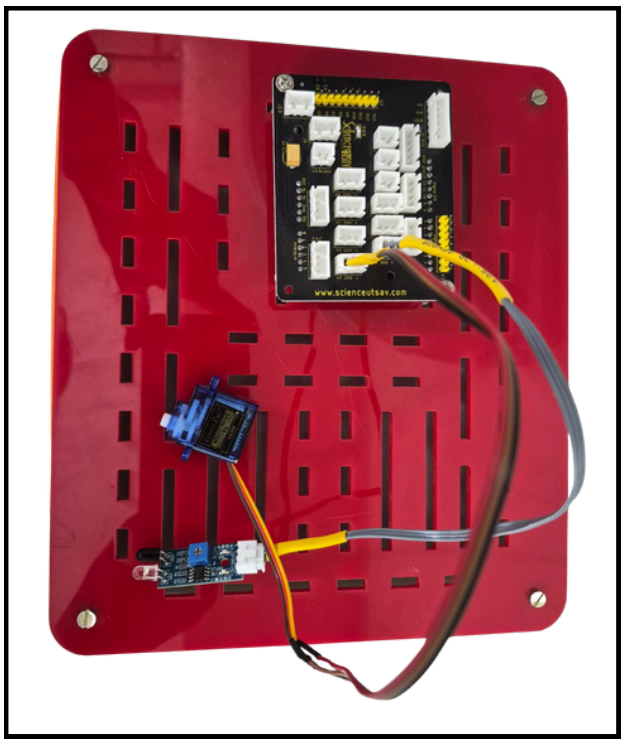
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

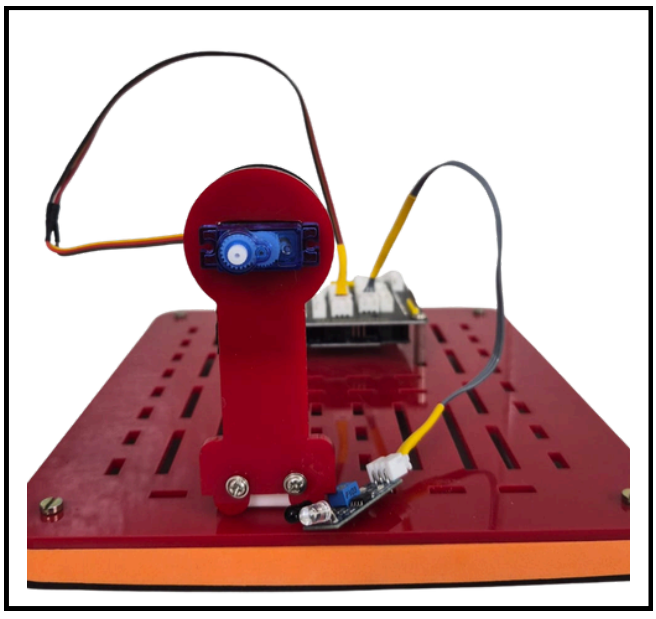
Step 1- Connect the IR sensor to the board



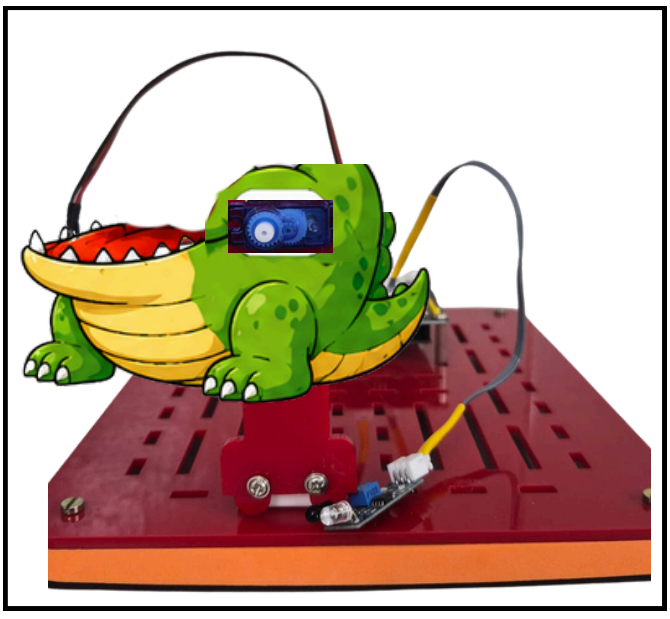
Step 2- Attach the servo motor to the board



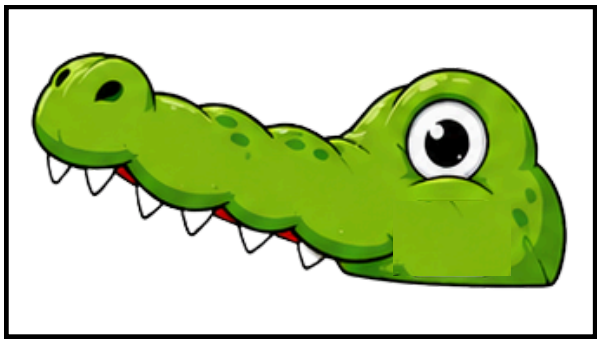
Step 3- Attach a stand to the base and insert the servo motor in it



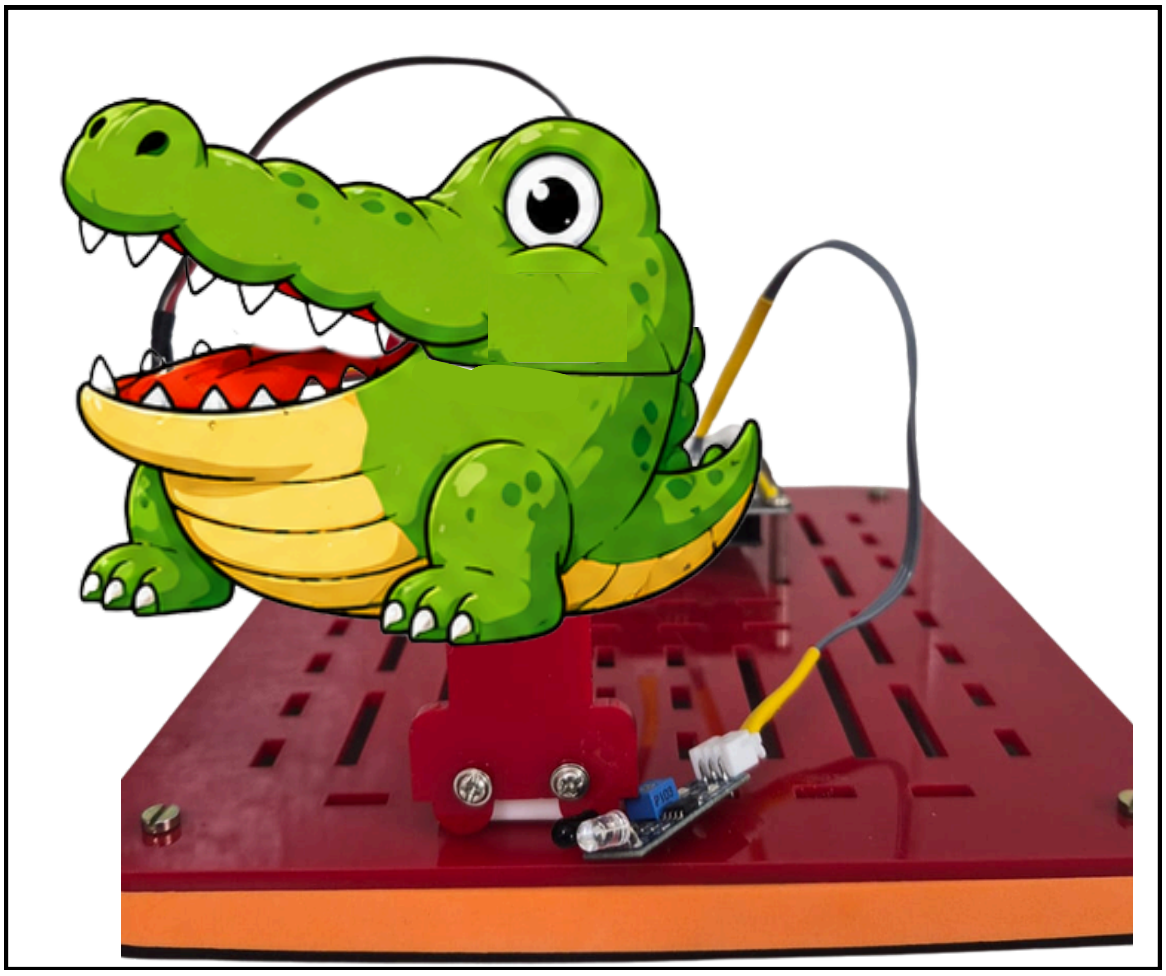
Step 4- Screw in the template



Step 5- Insert the upper template



Step 6- Upload the code and observe

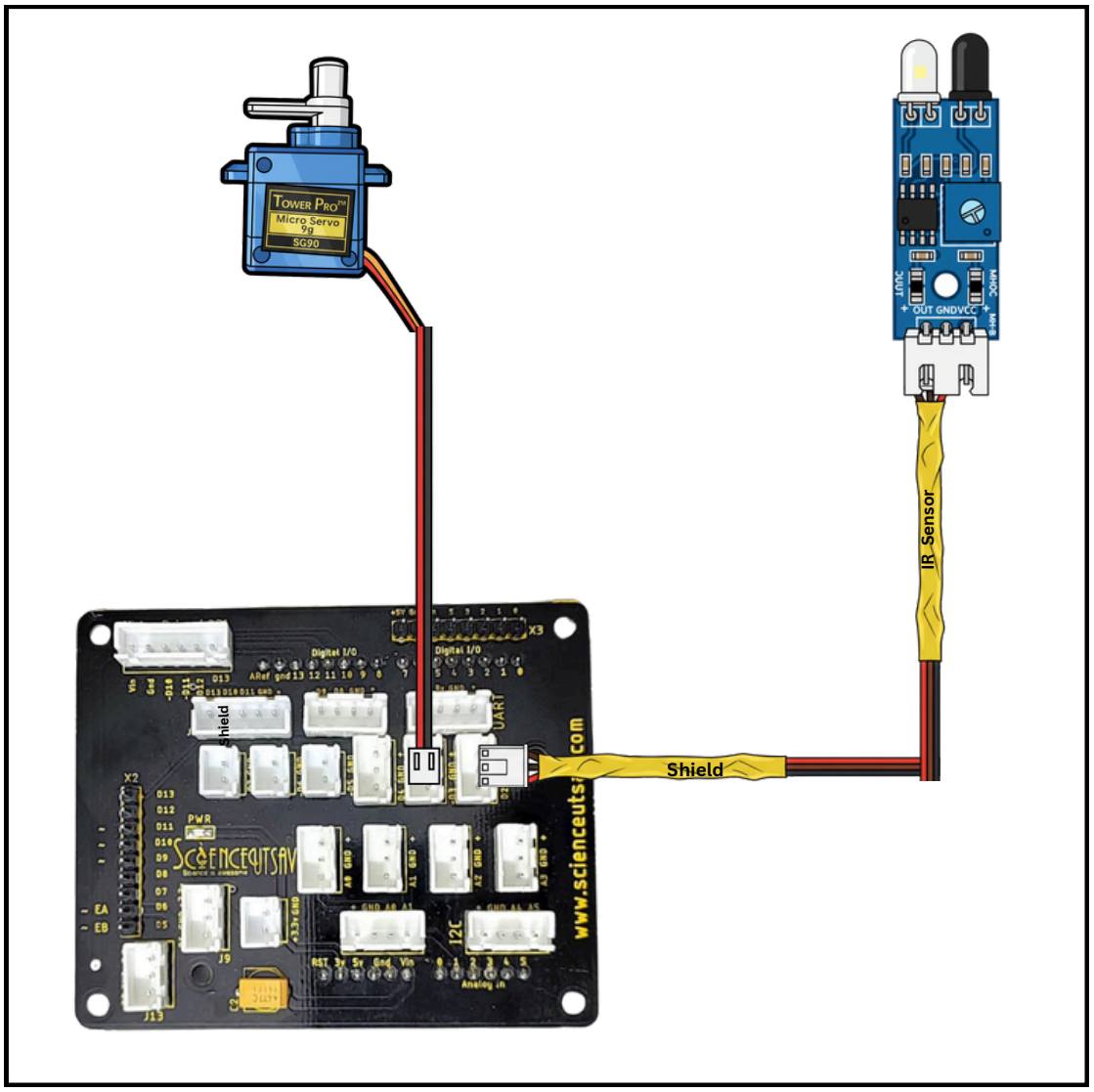


Code

```
// Session 6 Grade 6 Project 2
 // Crocodile Face Movement
 // IR detected = crocodile face moves

 #include <Servo.h>

 const int irSensor = 2;
 const int servoPin = 3;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 Servo faceServo;

 void setup() {
 pinMode(irSensor, INPUT);
 faceServo.attach(servoPin);
 faceServo.write(0);
 }

 void loop() {
 if (digitalRead(irSensor) == IR_DETECTED) {
   faceServo.write(90);
   delay(1000);
   faceServo.write(0);
   delay(500);
 }
 }
```

INTERMEDIATE PROJECT

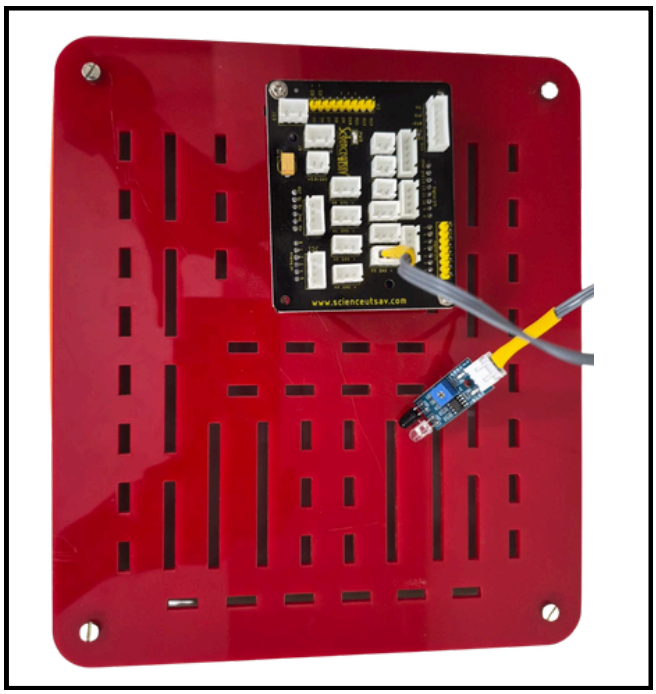
The circuit diagram of the project is given below



CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

Step 1- Connect the IR sensor to the board



Step 2- Attach the servo motor to the board

Step 3- Attach the stand vertically and the sevomotor holder horizontally as shown

Step 4- Attach the straw to the other template and fix the shaft to the straw

Step 5- Attach the template to the stand as shown

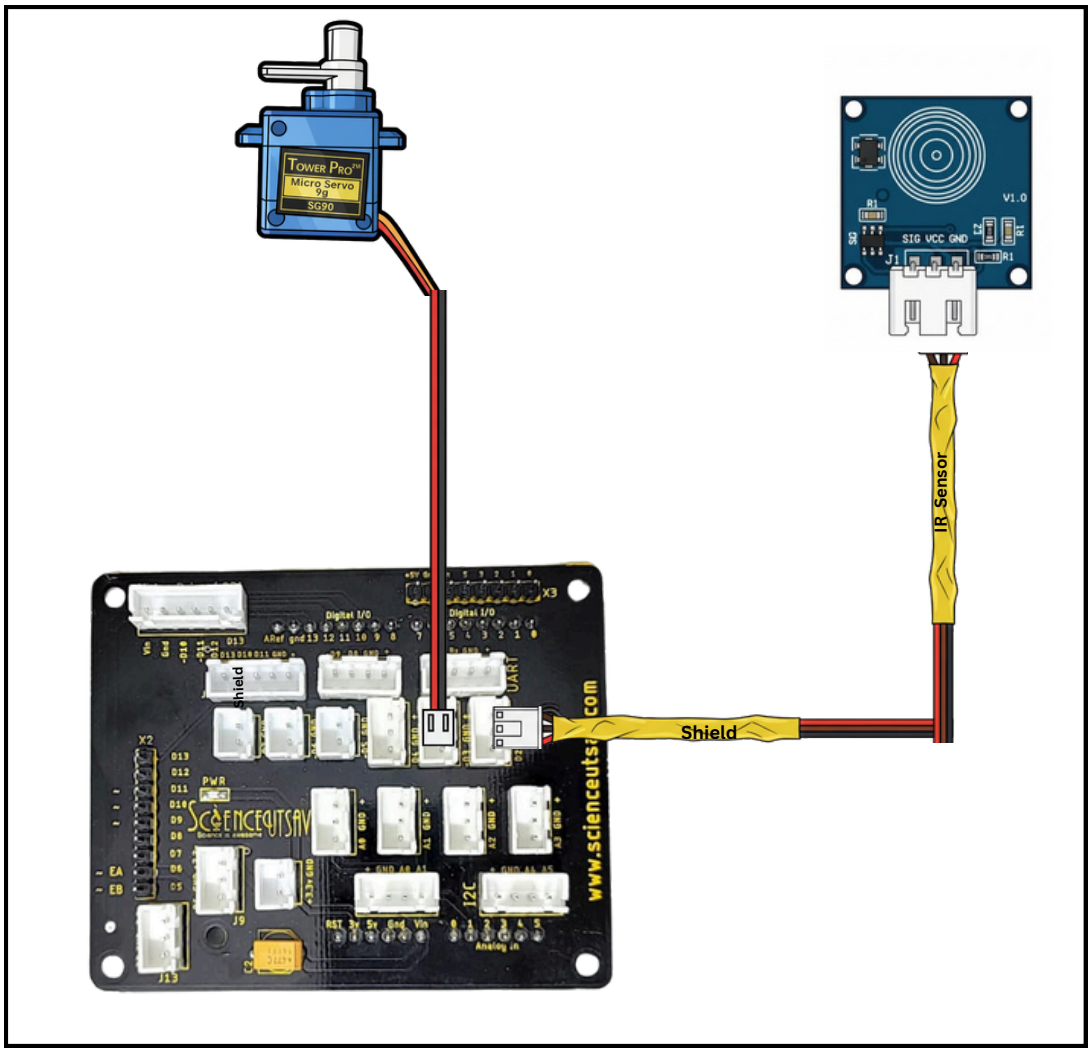
Step 6- Fix the shaft to the servomotor

Code

```
// Session 6 Grade 7 Project 1
 // Automatic Door
 // IR detected = door opens for 5 seconds and closes

 #include <Servo.h>

 const int irSensor = 2;
 const int servoPin = 3;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 Servo doorServo;

 void setup() {
 pinMode(irSensor, INPUT);
 doorServo.attach(servoPin);
 doorServo.write(0);         // door closed
 }

 void loop() {
 if (digitalRead(irSensor) == IR_DETECTED) {
   doorServo.write(90);      // door open
   delay(5000);
   doorServo.write(0);       // door close
   delay(500);
 }
 }
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below



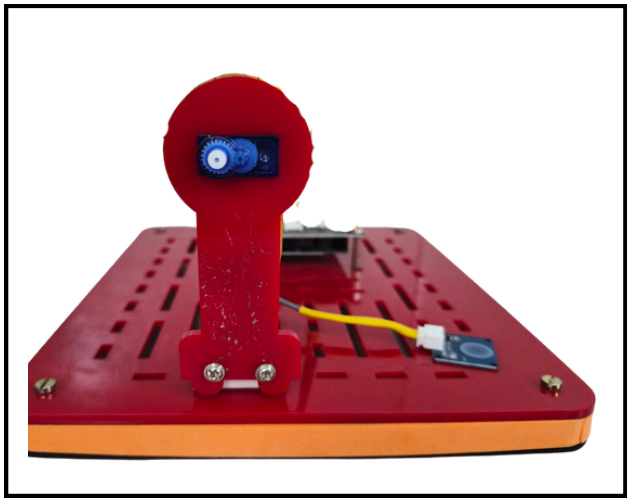
CONNECTION:

1. Touch sensor- D2
2. Servomotor- D3

Step 1- Fix the touch sensor to the board

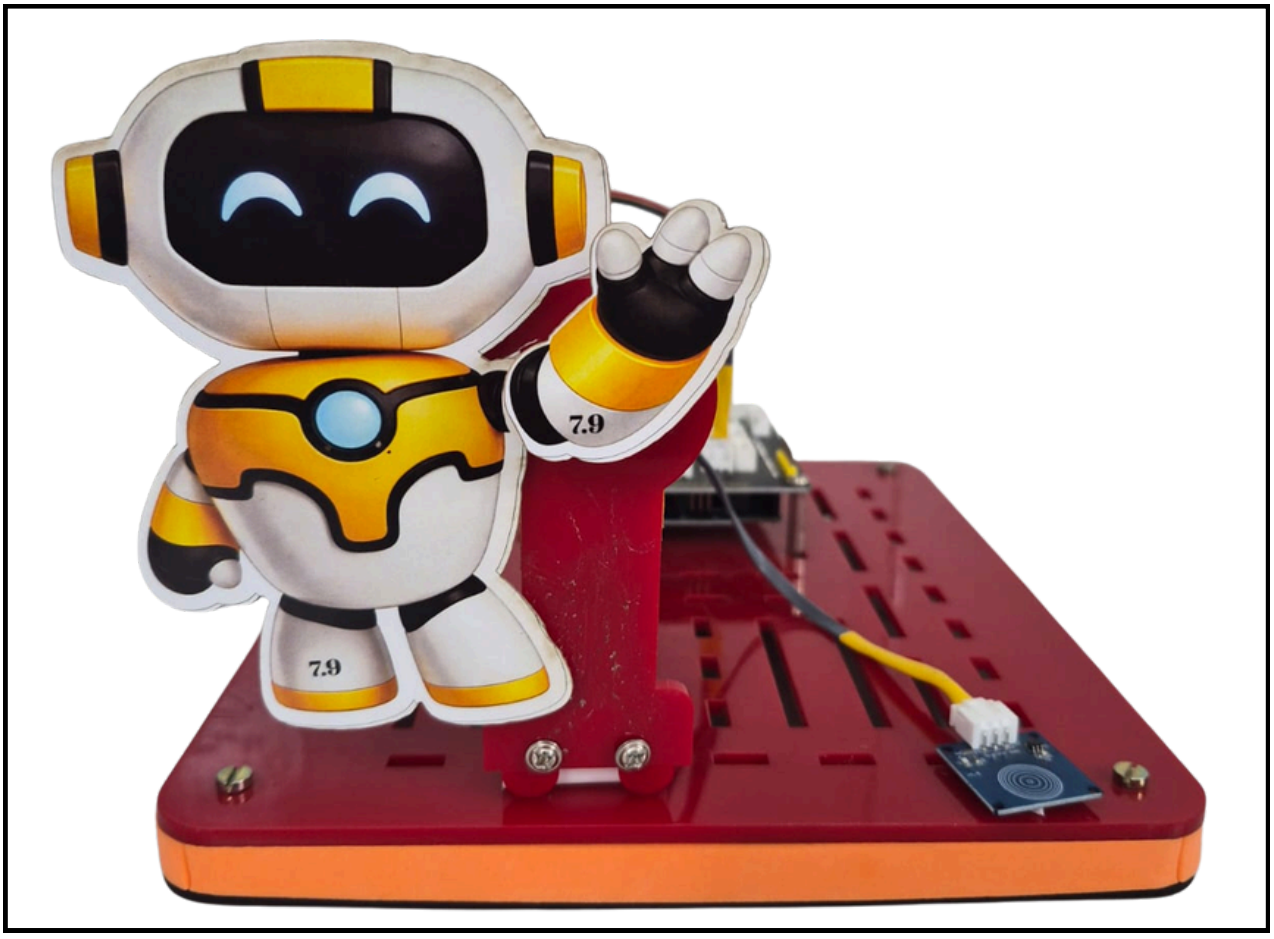
Step 2- Connect the servo motor as shown

Step 3- Fix the stand and insert the servo motor as shown



Step 4- Attach the template to the stand using bolts and attach the hand to the servo meter

Step 5- Upload the code and observe

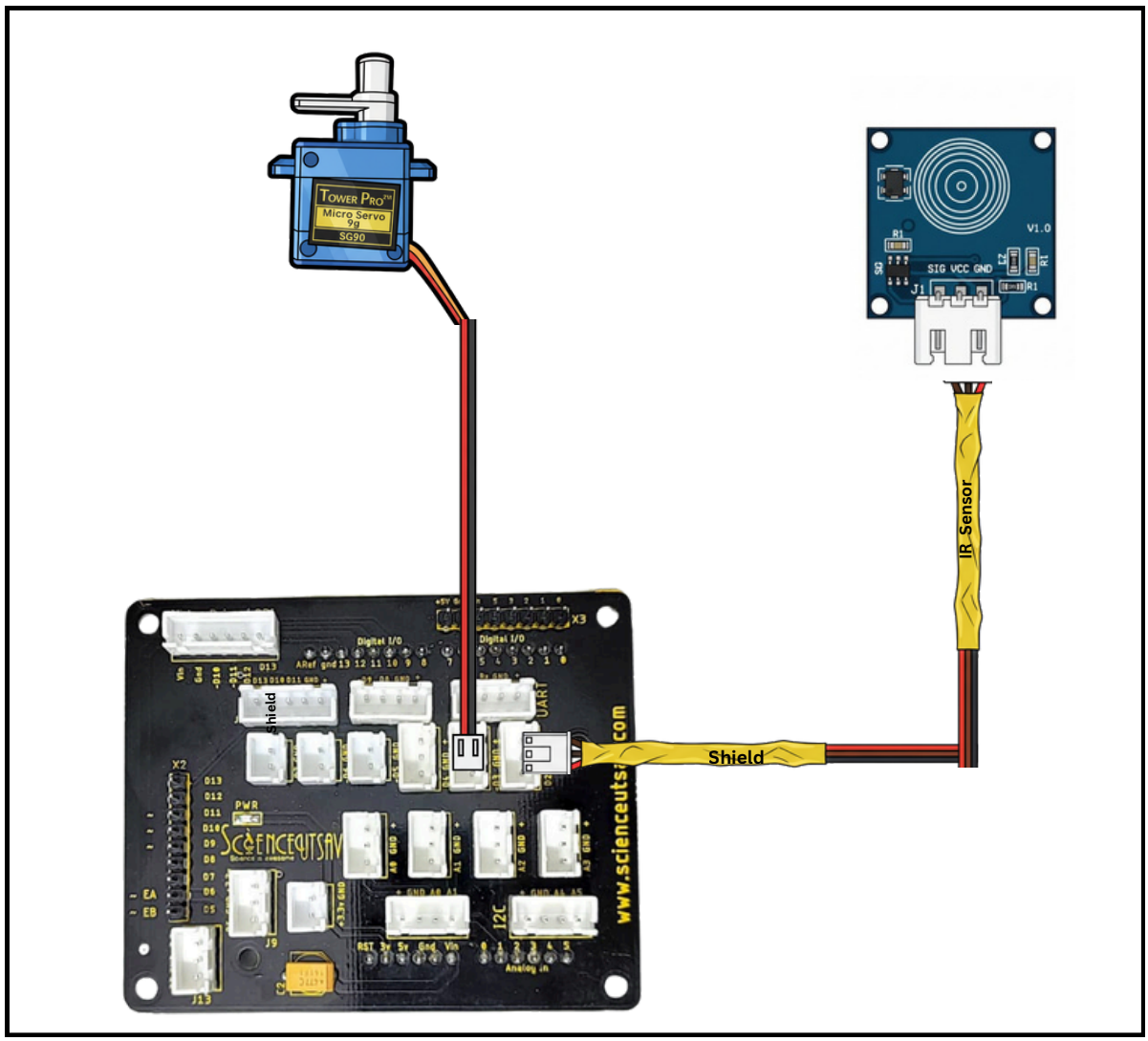


Code

```
// Session 6 Grade 7 Project 2
 // Robo Hand Hi
 // Touch detected = robot waves hand

 #include <Servo.h>

const int TOUCH_PIN = 2;
const int SERVO_PIN = 3;

Servo myServo;

void setup() {
  pinMode(TOUCH_PIN, INPUT);
  myServo.attach(SERVO_PIN);

  myServo.write(0); // Hand starts down/at rest
}
void loop() {
  int touchState = digitalRead(TOUCH_PIN);

  if (touchState == HIGH) {

    // The Waving Sequence (2 quick shakes)
    for (int i = 0; i < 2; i++) {
      myServo.write(75);    // Move up to wave (75° keeps it a bit tighter/faster than
90°)
      delay(200);           // Quick pause at the top

      myServo.write(15);    // Move back down
      delay(200);           // Quick pause at the bottom
    }

    myServo.write(0);       // Put the hand back to rest position
    delay(500);             // Cooldown so it doesn't immediately wave again if you linger
your finger
  }
}
```

ADVANCED PROJECT

The circuit diagram of the project is given below



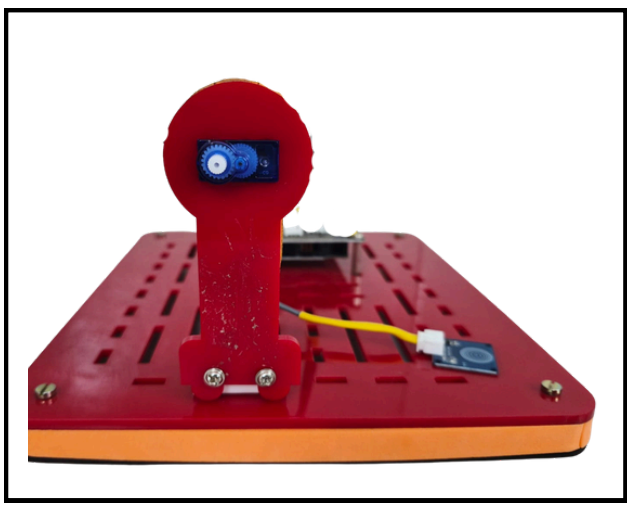
CONNECTION:

1. Touch sensor- D2
2. Servomotor- D3

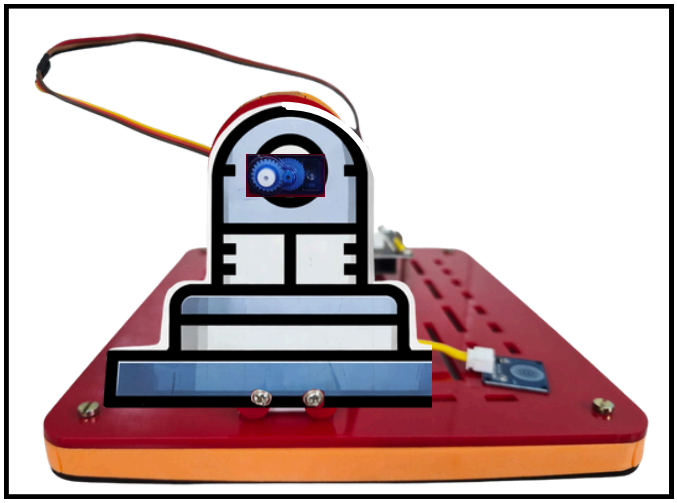
Step 1- Fix the touch sensor to the board

Step 2- Connect the servo motor as shown

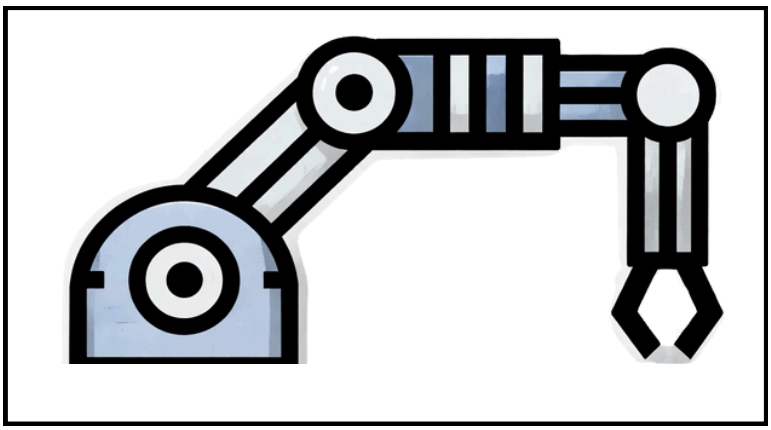
Step 3- Fix the stand and insert the servo motor as shown

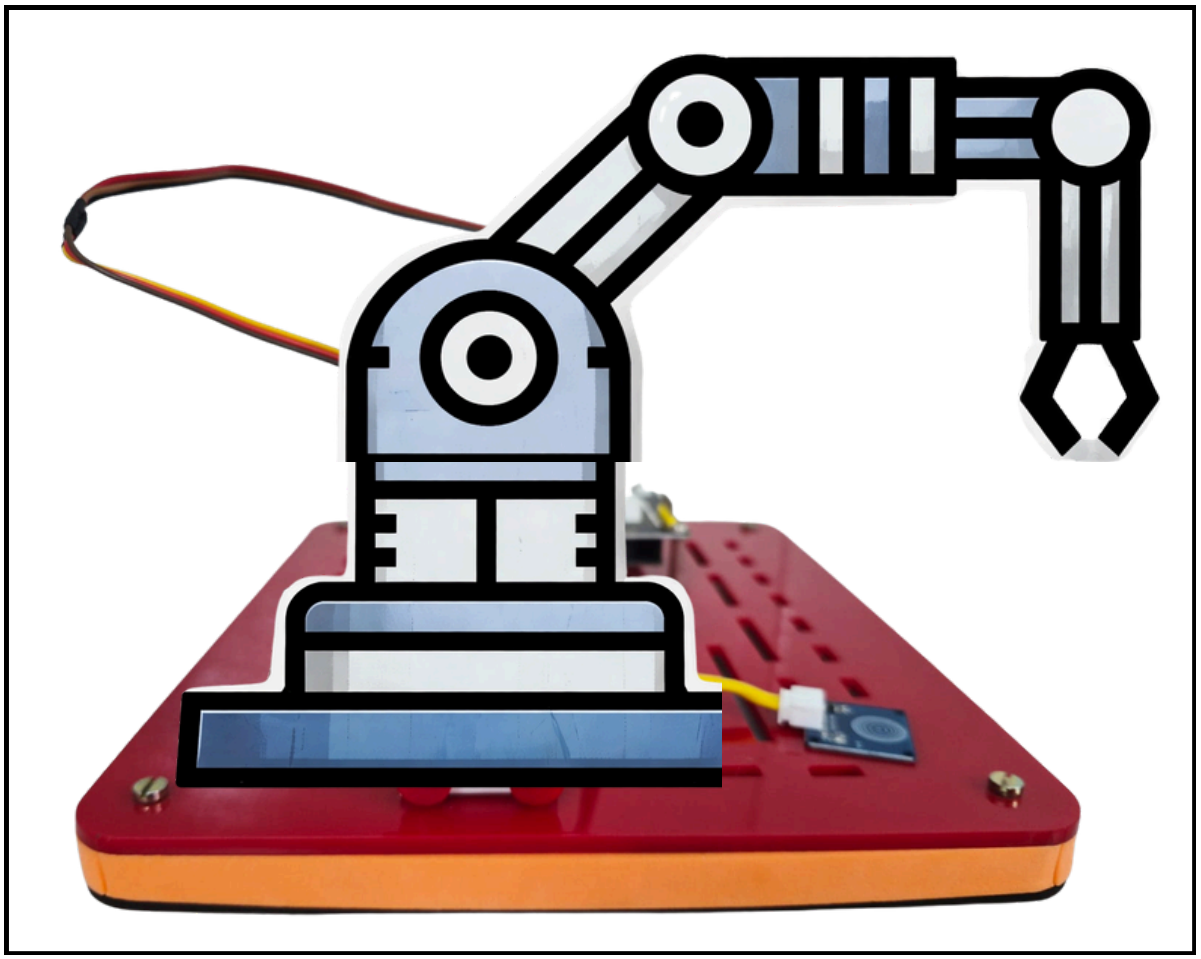


Step 4- Attach the template to the stand using bolts



Step 5- Attach the hand to the servo meter



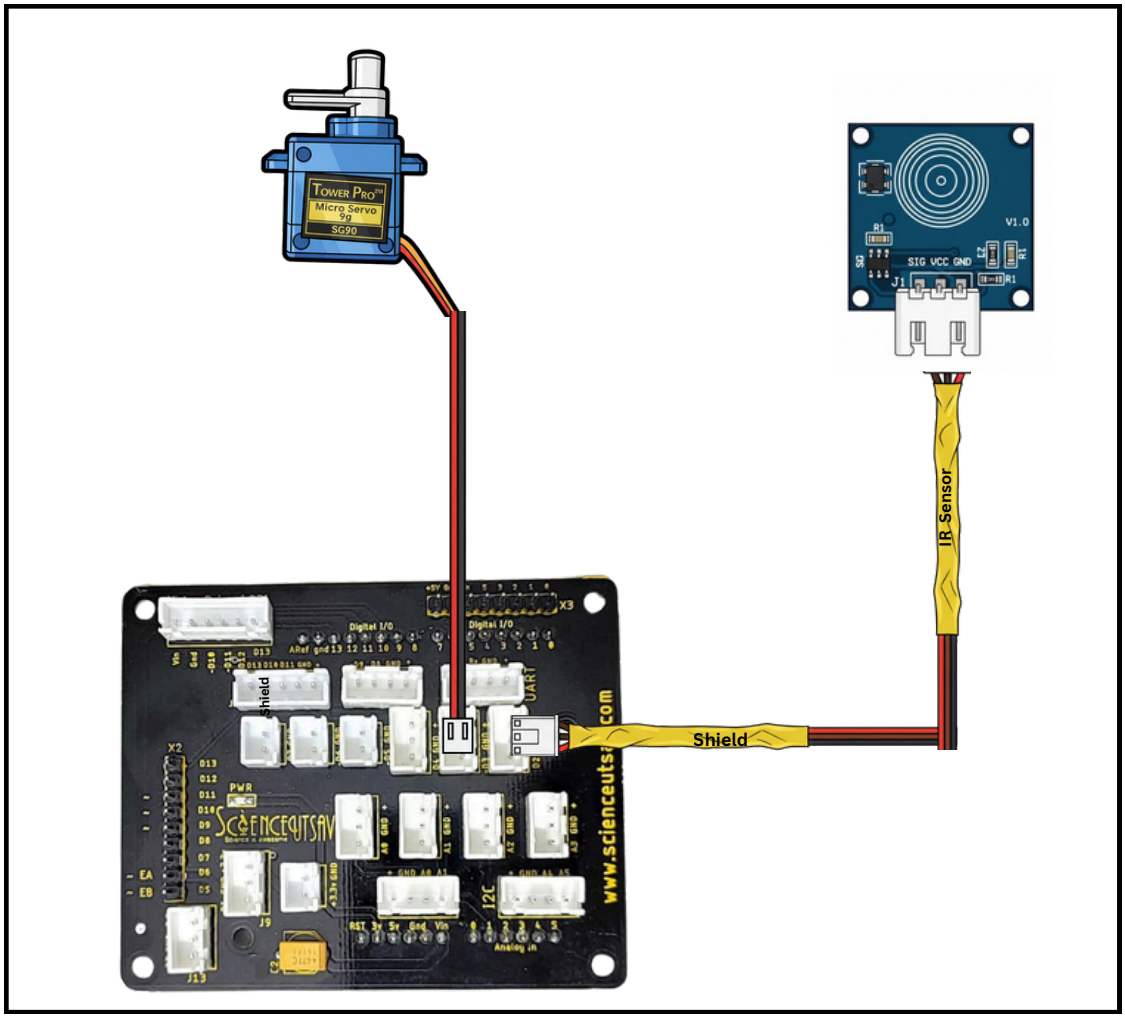


Code

```
// Session 6 Grade 8 Project 1
 // Crane Movement
 // Touch sensor touched = crane arm goes down and up

 #include <Servo.h>

 const int touchSensor = 2;
 const int servoPin = 3;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 Servo craneServo;

 void setup() {
 pinMode(touchSensor, INPUT);
 craneServo.attach(servoPin);
 craneServo.write(0);        // crane up
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
   craneServo.write(100);    // crane down
   delay(1000);
   craneServo.write(0);      // crane up
   delay(500);
 }
 }
```

ADVANCED PROJECT

The circuit diagram of the project is given below



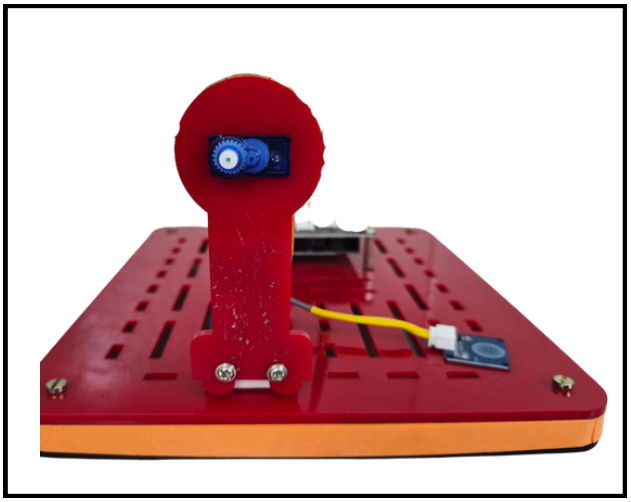
CONNECTION:

1. Touch sensor- D2
2. Servomotor- D3

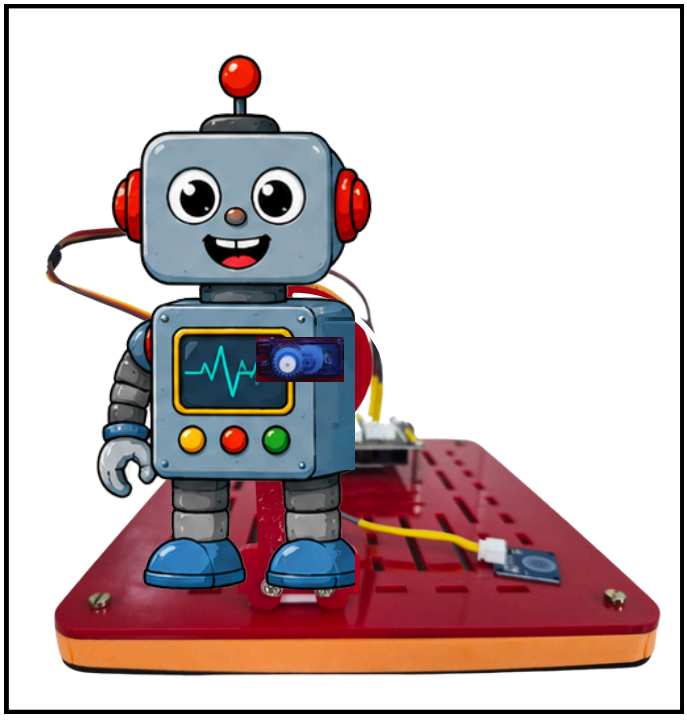
Step 1- Fix the touch sensor to the board

Step 2- Connect the servo motor as shown

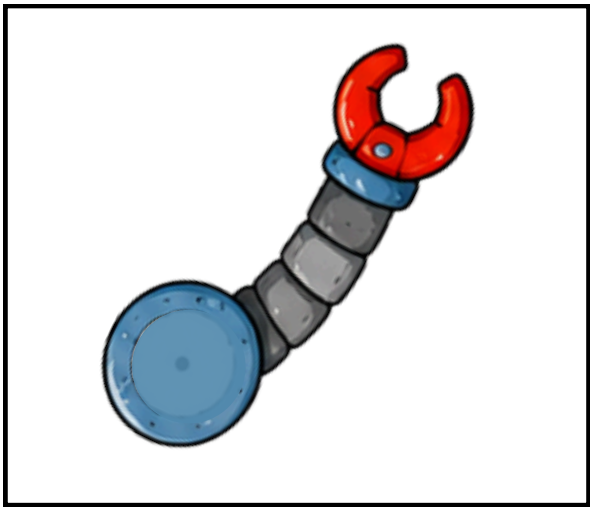
Step 3- Fix the stand and insert the servo motor as shown

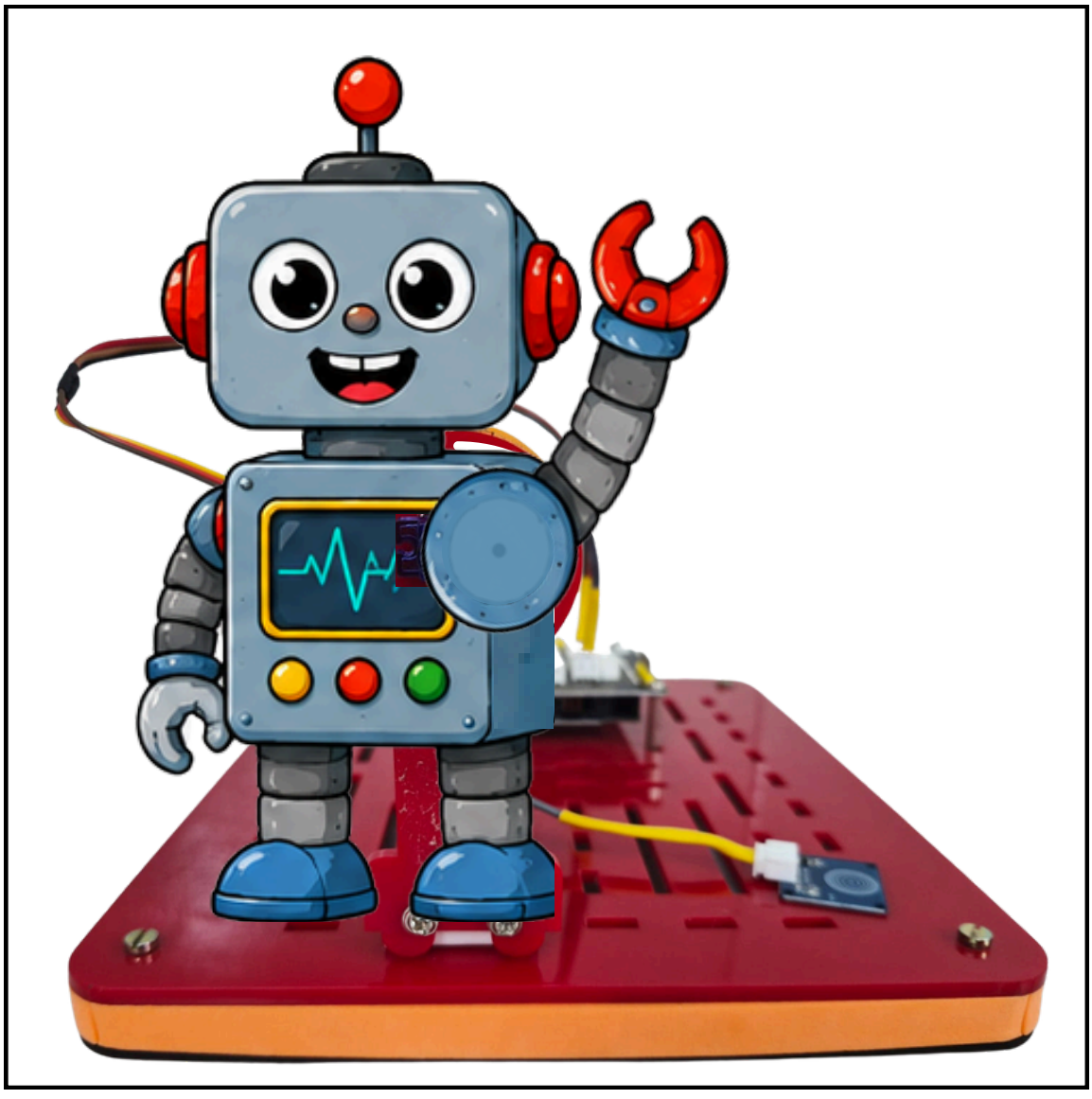


Step 4- Attach the template to the stand using bolts



Step 5- Attach the hand to the servo meter



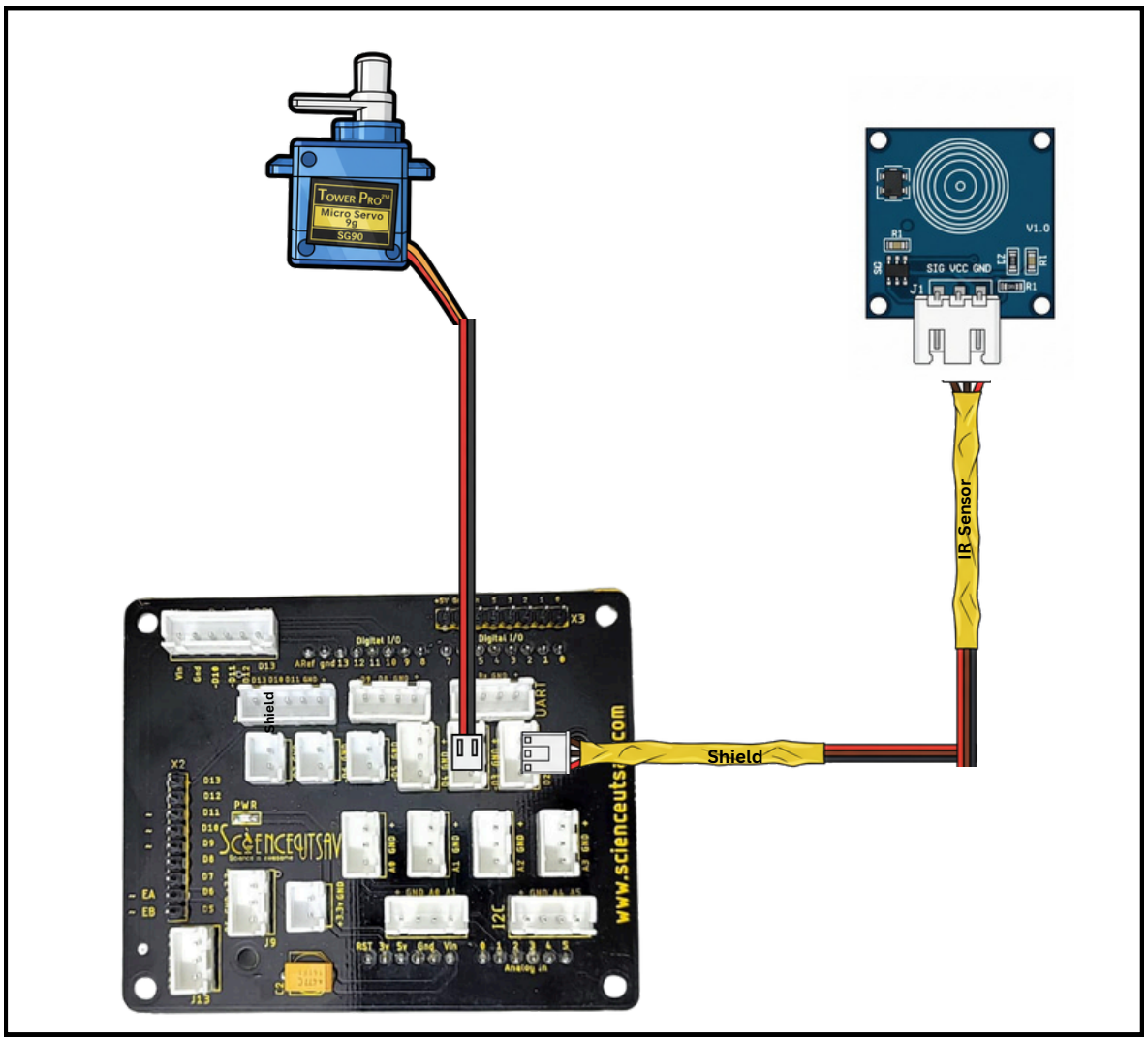


Code

```
// Session 6 Grade 8 Project 2
 // Robo Hi With Touch
 // Touch sensor touched = robot waves hand

 #include <Servo.h>

 const int touchSensor = 2;
 const int servoPin = 3;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 Servo handServo;

 void setup() {
 pinMode(touchSensor, INPUT);
 handServo.attach(servoPin);
 handServo.write(90);
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
   for (int i = 0; i < 3; i++) {
     handServo.write(30);
     delay(400);
     handServo.write(120);
     delay(400);
   }
   handServo.write(90);
   delay(500);
 }
 }
```

EXPERT PROJECT

The circuit diagram of the project is given below



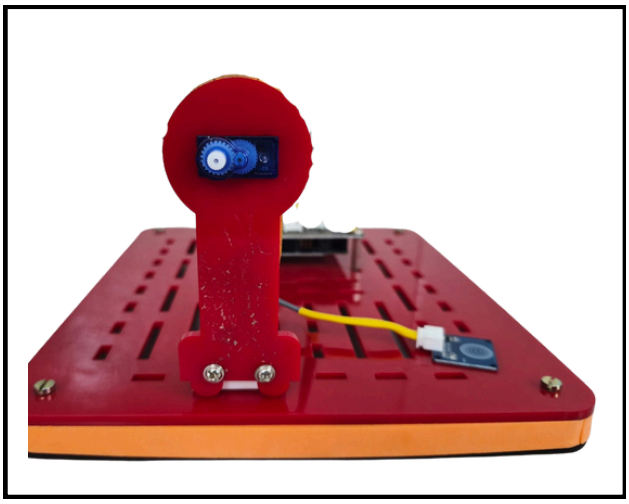
CONNECTION:

1. Touch sensor- D2
2. Servomotor- D3

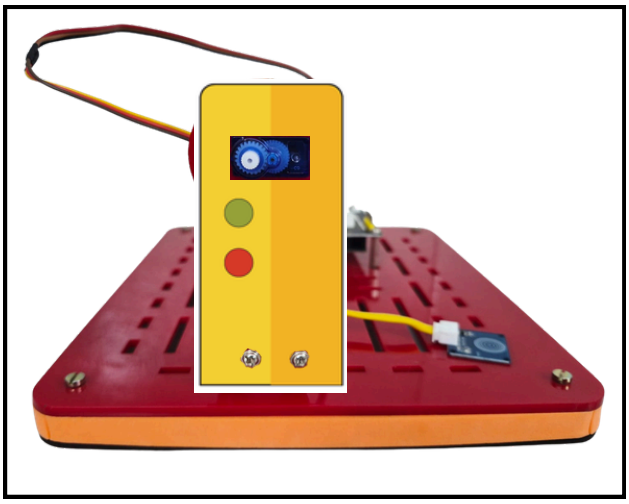
Step 1- Fix the touch sensor to the board

Step 2- Connect the servo motor as shown

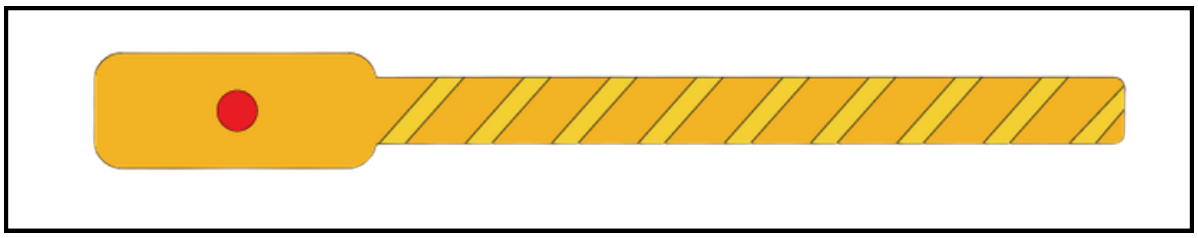
Step 3- Fix the stand and insert the servo motor as shown

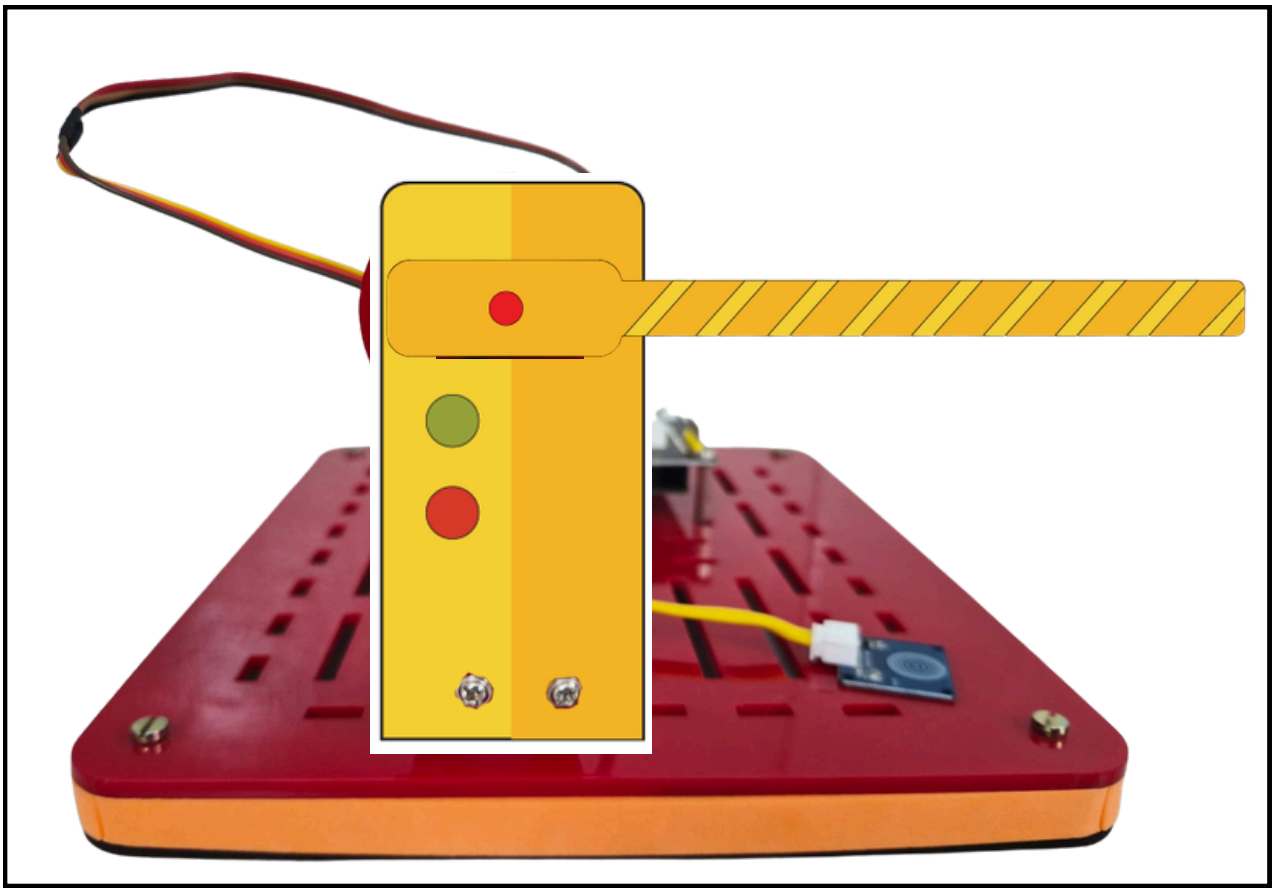


Step 4- Attach the template to the stand using bolts



Step 5- Attach the toll template to the servo motor



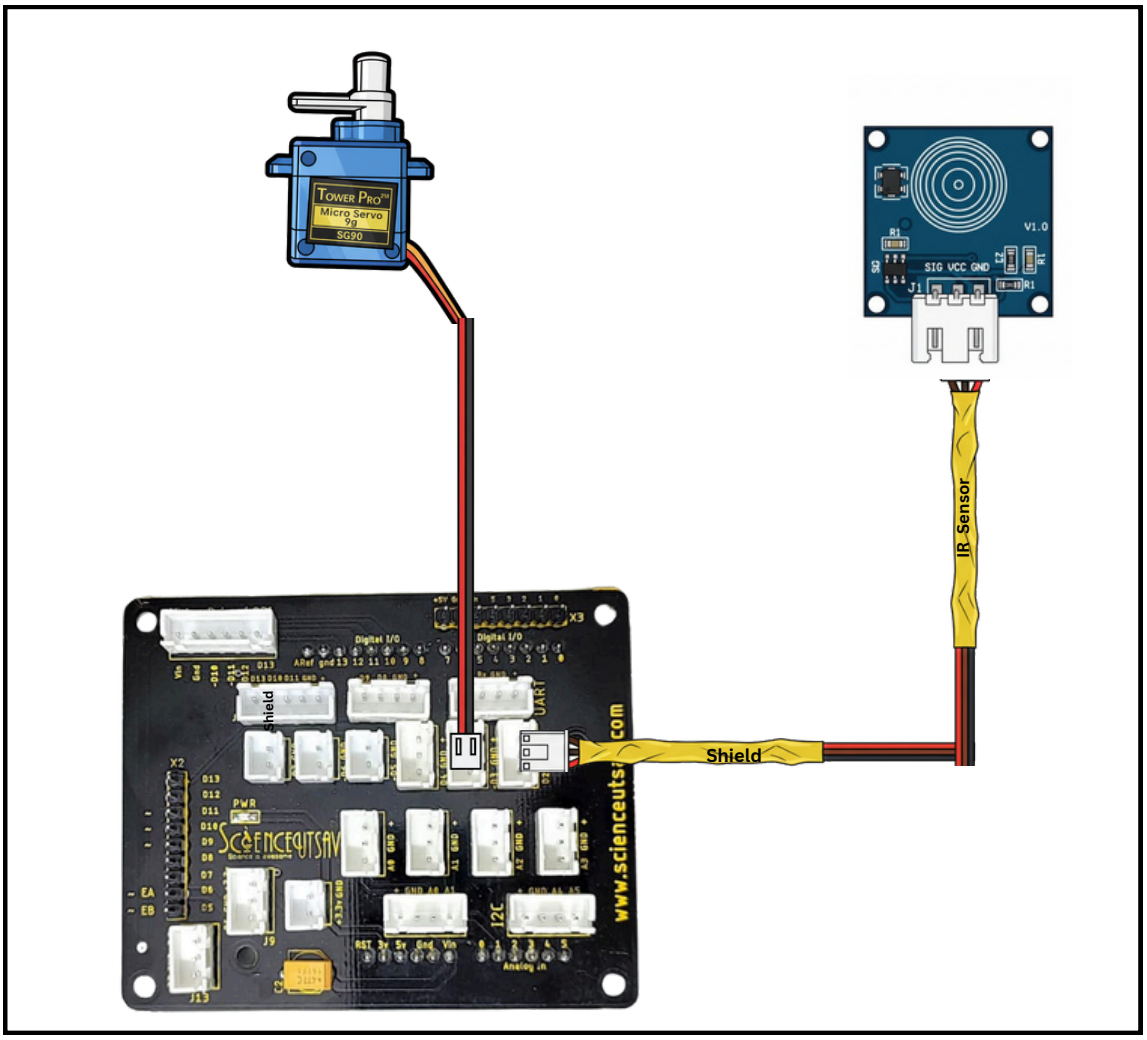


Code

```
// Session 6 Grade 9 Project 1
 // Touch Servo Gate
 // Touch sensor touched = gate opens for 5 seconds

 #include <Servo.h>

 const int touchSensor = 2;
 const int servoPin = 3;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 Servo gateServo;

 void setup() {
 pinMode(touchSensor, INPUT);
 gateServo.attach(servoPin);
 gateServo.write(0);         // gate closed
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
   gateServo.write(90);      // gate open
   delay(5000);
   gateServo.write(0);       // gate close
   delay(500);
 }
 }
```

EXPERT PROJECT

The circuit diagram of the project is given below



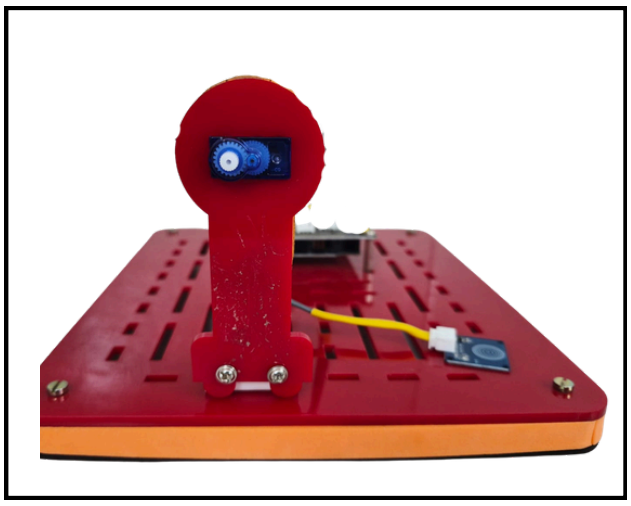
CONNECTION:

1. Touch sensor- D2
2. Servomotor- D3

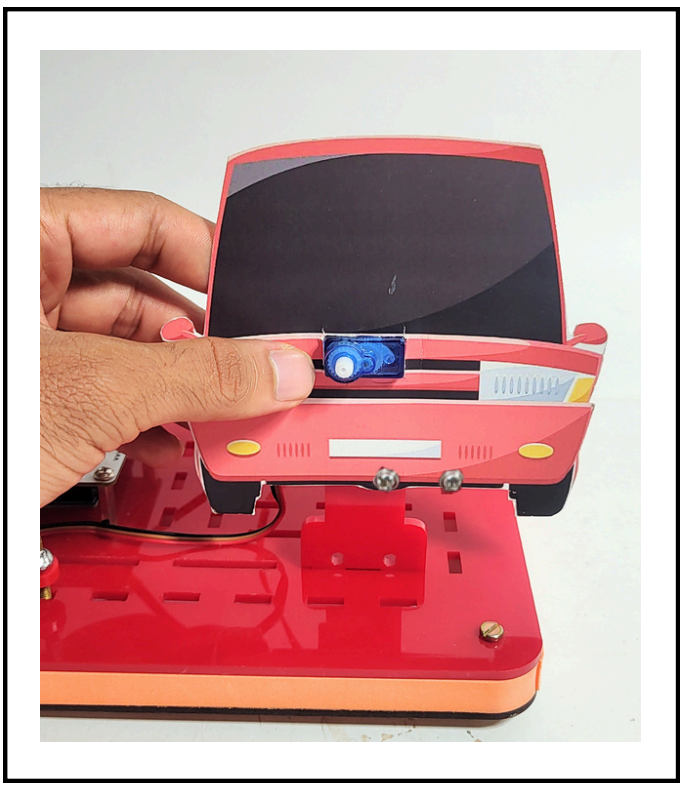
Step 1- Fix the touch sensor to the board

Step 2- Connect the servo motor as shown

Step 3- Fix the stand and insert the servo motor as shown

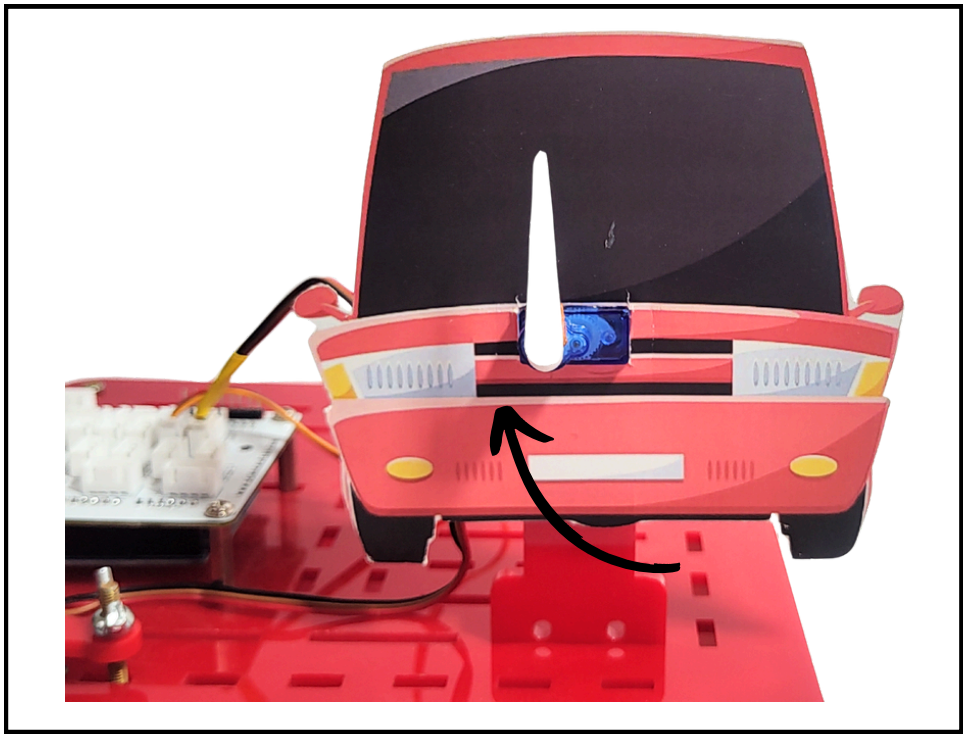


Step 4- Attach the template to the stand using bolts



Step 5- Connect the servo motor shaft

Step 6- Attach the wiper template and run the code



Code

```
// Session 6 Grade 9 Project 2
 // Car Wiper
 // Touch sensor touched = wiper moves left and right for 5 seconds

 #include <Servo.h>

 const int touchSensor = 2;
 const int servoPin = 3;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 Servo wiperServo;

 void setup() {
 pinMode(touchSensor, INPUT);
 wiperServo.attach(servoPin);
 wiperServo.write(90);
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
   unsigned long startTime = millis();

   while (millis() - startTime < 5000) {
     wiperServo.write(30);
     delay(300);
     wiperServo.write(150);
     delay(300);
   }

   wiperServo.write(90);
 }
 }
```